

MEET NAKUM

AI Engineer | GenAI Developer | Autonomous Systems Developer

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Professional Summary

AI Engineer and M.Tech (Artificial Intelligence & Data Science) candidate with experience building autonomous AI agents, multimodal assistants, LLM-powered systems, AI security frameworks, and intelligent automation platforms. Developed NYRA, a locally deployed AI assistant featuring voice interaction, screen understanding, memory retention, system control, and autonomous task execution – designed with a modular architecture for future cross-platform expansion. Strong expertise in Agentic AI, Generative AI, Prompt Engineering, Autonomous Systems, LLM Security, Computer Vision, and Human-AI Interaction.

Technical Skills

Programming Languages: Python, SQL, C, JavaScript

AI / ML: Machine Learning, Deep Learning, Generative AI, Agentic AI, Prompt Engineering, LLM Application Development, Retrieval-Augmented Generation (RAG), NLP, Autonomous AI Systems

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-Learn, LangChain, Hugging Face Transformers, Ollama, OpenAI SDK, Gemini API

Computer Vision: OpenCV, YOLO, Image Processing

Databases: SQLite, ChromaDB, FAISS, Vector Databases

Tools & Technologies: Git, GitHub, Docker, Linux, FastAPI, Playwright, PyQt6, REST APIs

Projects

NYRA – Autonomous Personal AI Assistant

GitHub

- Built a locally deployed autonomous AI assistant on Linux, with a modular architecture designed for future Windows and macOS compatibility.
- Implemented real-time voice interaction, wake-word detection, screen understanding, memory retention, and intelligent task automation.
- Built multimodal capabilities including file analysis, web research, computer control, contextual reasoning, and workflow execution.
- Integrated browser automation and OS-level controls using Python and Playwright, enabling end-to-end task execution without human intervention.
- Designed a PyQt6 holographic interface featuring multiple voice-reactive visualization modes.
- Developed a persistent memory architecture enabling personalized long-term interactions and contextual task execution across sessions.
- Combined local and cloud LLMs (Ollama + OpenAI/Gemini) to enable autonomous reasoning and decision-making.

AEGIS V – Autonomous AI Immunity Framework

GitHub

- Designed a bio-inspired multi-layer defense architecture for protecting Large Language Models against jail-break, prompt injection, adversarial, and social engineering attacks.
- Developed Cognitive Membrane, Intent Tracker, Self-Hardening Core, and Blockchain Audit modules for adaptive threat detection.
- Implemented vector-based threat analysis using embeddings and cosine similarity for real-time attack classification.
- Built an autonomous learning mechanism capable of generating new threat signatures without retraining the underlying model.
- Evaluated the framework on 5,500+ prompts spanning multiple attack categories; achieved **95.27% classification accuracy**, **97.88% recall**, and **95.39% F1-score**.

Publications

Comprehensive Analysis of Air Quality Trends in India Using Machine Learning and Deep Learning Models

ICDCN 2025 | ACM International Conference on Distributed Computing and Networking

- Conducted comparative analysis of Deep Learning and Machine Learning models for PM2.5 forecasting across Indian states.
- Evaluated RNN, LSTM, GRU, ARIMA, and SARIMA models; achieved **~99% prediction accuracy** using RNN-based forecasting.
- Demonstrated effectiveness of deep learning approaches for large-scale environmental time-series forecasting.

A Real-Time Geolocation Framework for Enhanced Medicine Accessibility: Integrating Flask, SQLite, and OpenStreetMap in Urban Healthcare Systems

IRJMETS | April 2025

- Designed and implemented a real-time medicine search and pharmacy discovery platform using Flask REST APIs, SQLite, and OpenStreetMap.
- Achieved **150ms average request latency** and **100% geospatial accuracy** using Haversine-based distance calculations.
- Enabled real-time medicine availability notifications and bidirectional pharmacy-user communication.

Education

DGU DBS Global University, Dehradun 2025 – 2027

M.Tech in Artificial Intelligence & Data Science CGPA: 8.99 (S1: 8.80 | S2: 9.17)

Jain University, Bengaluru 2021 – 2025

B.Tech in Computer Science & Engineering CGPA: 7.60

Experience

Data Analyst Intern (Remote) – Unified Mentors Jan 2024 – Mar 2024

- Performed data cleaning, transformation, and exploratory data analysis on 3+ real-world datasets using Python and Pandas.
- Built analytical dashboards and reports to support data-driven decision-making across business teams.
- Automated reporting workflows, reducing manual data processing time and improving pipeline efficiency.
- Delivered analysis across Amazon Sales Analytics, Bird Strike Visualization, and Agricultural Data Analysis projects.

Certifications

- Data Analytics and Visualization Job Simulation – Forage
- Python 101 for Data Science – IBM / Cognitive Class
- Data Structures in C – Udemy

Research Interests

Agentic AI, Generative AI, Autonomous Systems, LLM Security, Computer Vision, Human-AI Interaction, Multimodal AI Systems, NLP.